Replication Package for: A Welfare Analysis of Policies Impacting Climate Change

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October 16, 2025

1 Overview

The replication package allows anyone to re-create all figures, tables, and numbers reported in the text using Stata. There is also a series of input assumptions that users can change to generate new MVPF estimates. The replication package is structured as follows. Section 2 will contain the Data Availability Statement. Section 3 will list the necessary computational requirements for running the replication packet. Section 4 will outline the folder structure for the code and data used for replication and will discuss the key code files and data files in greater detail. Section 5 will detail the steps for running the replication package and includes instructions for changing the code to reflect alternative specifications. The code takes significant time (> 2 days) to run the entire replication package, including generating all the estimates with bootstrapped standard errors.

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2 Data Availability Statement

Datasets and cites for data sources used in the analysis are provided in the following locations:

- data/1_assumptions/policy_category_assumptions_MASTER.xlsx
- data/1_assumptions/evs/Record of EV Data Sources.xlsx
- data/1_assumptions/default_assumptions_toggles_vMAIN.xlsx
- all files in data/2a_causal_estimates_papers
- all files in data/1_assumptions/evs
- all files in data/1_assumptions/marginal_damages

All data are publicly available and provided in the replication packet.

3 Computational Requirements

3.1 Software Requirements

The replication package requires the following programs to be installed:

- Stata (code was last run with version 19.5)
- Matlab and Mathematica

3.2 Controlled Randomness

Random seed is set on line 74 in wrapper_metafile.do.

3.3 Memory, Runtime, and Storage Requirements

- Approximate time needed to reproduce the analyses on a standard 2025 desktop machine: 80 Hours
- Approximate storage space needed: 220 MB

Details: The code was last run on a 2021 Macbook Pro (M1 Max) with 64GB of RAM.

4 Description of programs/code

This list describes all of the folders besides the data folder.

4.1 Code Structure

- masterfile.do
  - Wrapper that runs through the code files below, generating all figures, tables, and in-text numbers.
- ado
  - Stores all of the ado files, which wrapper/macros.do calls in except the cost_curve_masterfile.ado
- bootstrapping
  - Stores the main file that implements bootstrapping.do along with all the files called in bootstrapping.do
- calculations
  - Stores pre-processing files called in macros.do to construct environmental externalities for each policy
- cost_curve
  - Stores the ado file used to calculate cost curve externalities
- data_cleaning
  - Stores pre-processing files used to clean electric vehicle-related data
- figtab
  - Stores all the do files that are needed to create the tables and figures in the paper
- policies
  - The Harmonized subfolder stores all of the policy do-files the main MVPF sample uses
  - The Robustness subfolder stores policy do-files only used for robustness analysis
- publication_bias
  - Stores all the do files that are needed to create the publication bias-corrected version of the estimates

- wrapper
  - Stores the main do files called in masterfile.do, including macros.do, metafile.do, and prepare_causal_estimates.do
  - Stores the do files used to create all the main and appendix figures and tables
  - Stores robustness.do, which creates all the robustness numbers reported in the text

4.2 Explanation of Important Code Files

Macros.do This file, stored in the wrapper subfolder, calculates all of the externalities used for each policy's MVPF. Therefore, any change in specifications requires the user to re-run macros.do. If you are running through masterfile.do, it will automatically call macros.do. Macros.do creates the electricity, natural gas, and vehicle-related externalities.

Policy do files In the code/policies/harmonized subfolder, there is a separate do file for each policy used in the analysis. For a mapping of the policy do-file name to the policy name reported in Table One, see files/policy_details_v3.xlsx.

You cannot run each policy do-file alone. Instead, the masterfile must call it. To run an individual policy, a user can change the all_programs positional argument in the masterfile to the name of the specific policy. Each policy loads in the causal effect estimates from the relevant paper(s). These may be the adjusted causal estimates when run externally for bootstrapping or publication bias. Next, the policy do-files calculate each policy's WTP and net government cost. The do-files call on globals, datasets, and ado files introduced in macros.do to do this. Finally, the last section computes the MVPF as the ratio of the WTP and cost and stores all the relevant MVPF components as globals. Many do-files also store components you can use to create policy waterfall graphs if desired.

Prepare Causal Estimates Before estimating MVPFs and bootstrapping them, we prepare bootstrap draws of the causal estimates that go into each MVPF calculation. The file prepare_causal_estimates.do does this by looping over all the policies in our sample. For each policy, it finds the corresponding causal estimates file in causal_estimates/uncorrected, which details point estimates and standard errors. (In cases where standard errors are unavailable, we use t-statistics or p-values to back them out. Where the paper only reports significance levels, we draw uniformly over the range of possible associated p-values. We record the standard error as 0 in cases where the authors do not report anything. We do not use policies with missing standard errors when calculating standard errors.)

The file details the assumed correlational structure between estimates in the column "corr_direction." This variable defines blocks in the correlation matrix between estimates: we indicate blocks by different base numbers, and the sign determines the correlation direction. E.g., if we have four variables with corr_directions 1, -1, 2, 2 respectively, then 1 and 2 are perfectly negatively correlated, 3 and 4 are perfectly positively correlated, and 1 and 2 are uncorrelated with 3 and 4. We set these correlations to maximize the width of our confidence intervals where estimates are from the same sample. Given point estimates, standard errors, and a correlation matrix, we draw from the implied joint normal distribution 1000 times, generating the 1000 sets of estimates corresponding to our 1000 bootstrap estimates of the MVPF.

Bootstrapping To get confidence intervals for the MVPF estimates, we bootstrap standard errors. This process occurs in bootstrapping.do. Since learning-by-doing takes significantly more run time, we treat policies with and without learning-by-doing separately. For policies without learning-by-doing, we run the metafile with 1000 replications. The causal estimates are created in prepare_causal_estimates and fed into bootstrap_wrapper. For policies with learning-by-doing, we loop through a series of causal estimates to understand the functional form between the causal estimate and the resulting MVPF. Once we have estimated this functional form, we use it to create the bootstrapped MVPFs without running the metafile on 1000 replications. This entire process occurs in bootstrapping.do.

Publication Bias To estimate MVPFs corrected for publication bias, we run pub_bias_wrapper, which calls four nested do-files. prep_and_run_matlab estimates the level of publication bias in our sample of environmental policies. cdf_plot generates the implied CDFs from our method of estimating publication bias and from that of Andrews and Kasy (2019), both compared to the empirical CDF of the (absolute value of the) t-stats in our data. heuristic_graphs generates the "funnel plot" of the standard errors for the studies in our sample against the corresponding point estimates. Lastly, replace_w_corrected corrects the point estimates of the papers in our sample and edits Excel sheets that the code uses in MVPF calculations.

Robustness This file, stored in the wrapper subfolder, calculates the numbers reported in the text that we do not directly take from a table or figure. These numbers are outputted to the 4_results subfolder and saved as robustness.xlsx.

Figure & Table Creation We store the wrapper files that create all the tables and figures in the wrapper subfolder as tables.do, figures.do, appendix_tables.do, and appendix_figures.do. Each do-file has nested files in the figtab folder, which construct the individual figures and tables. All the tables and figures are stored in 5_graphs and 6_tables, respectively.

4.3 Data Structure

This list includes all the subfolders of the data folder.

- 0_log
  - After each run of the metafile, a new log file will be created and stored here
- 1_assumptions
  - Stores all assumptions used in the externality calculations.
- 2a_causal_estimates_papers
  - This file stores the causal estimates used in each policy analyzed. The name of the spreadsheet corresponds to the name of the policy do-file.
- 2b_causal_estimates_draws
  - This folder stores the bootstrapped causal estimates that are needed when calculating the bootstrapped standard errors. Each metafile run will produce a new subfolder named using the run's timestamp and the run's name. If the reps argument passed into the metafile is set to 0, the rest of the code will only use the point estimate of the causal estimate for each policy.
- 3_bootstrap_draws
  - Similar to 2b_causal_estimates_draws, each metafile run will produce a new subfolder that is named using the run's timestamp and the run's name. Instead of storing the bootstrapped causal estimates, this folder will store the bootstrapped MVPF components.
- 4_results
  - Similar to the folders above, each metafile run will produce a new subfolder in 4_results that is named using the run's timestamp and the run's name. This folder will have the main compiled results from the run. In other words, this file will store the MVPF along with the components used to construct the MVPF for each policy included in the run. If the same specification have been run more than once, the figure, table and robustness code are set up to use the most recent timestamp.
- 5_graphs
  - This folder stores the main and appendix figures reported in the paper. It also stores some processed data used to create the figures.
- 6_tables

    - This folder stores the main and appendix tables reported in the paper. It also stores some processed data that the code uses to create the tables.

- 7_publication_bias
    - Stores some data used to create the publication bias results.

4.4 Explanation of Important Data Files

Policy Assumptions Masterfile The assumptions that we use to calculate the environmental and fiscal externalities are primarily pulled from the 1_assumptions/policy_category_assumptions_MASTER.xlsx spreadsheet. We list the sources for the data pulled from the spreadsheet on the first tab.

Program Specific Causal Estimates The causal estimates used for each policy are stored in 2a_causal_estimates_papers/uncorrected_vJK. The names of the spreadsheets correspond to the name of each policy do-file. On the raw_data tab of each spreadsheet, we list where we are pulling the estimate in the paper, the link to the paper PDF, and the date we accessed it. We also show any transformations to the causal estimate before it enters the policy do-file.

5 Instructions to Replicators

To create all of the figures, tables, and numbers reported in the paper, the user only has to run the masterfile. In addition, prior to running alternative specifications or smaller sections of the code, the user must first run the masterfile all the way through. Once you run the masterfile, the results will populate in the subfolders of the data folder as described in Section 4.3. Directions are below:

5.1 Initial run/replicating all results

- Confirm that Mathematica and MATLAB has been downloaded to your computer.
- Open masterfile.do in the code folder.
- Make sure the three globals (rerun_data, bootstraps, run_pub_bias) on lines 14-16 are set to "yes".
- Set the user-specific file paths on lines 41-47:
    - Change the XX on lines 41 and 42 to the username where the file is stored on your computer.

- Make sure that the Dropbox and Github global on lines 43 and 44 point to the data folder and the general Github folder, respectively.
- If using a Mac, add the file location of Wolfram Alpha on line 46.

- Run masterfile.do from top to bottom.

5.2 Rerunning sections of the code/changing run specifications

- Recreating figures, tables, and robustness numbers:
  - Set the three globals on lines 14-16 to "no" and run the masterfile.
  - Figures, tables, and the robustness numbers are exported to the graphs subfolder, table subfolder, and to an Excel file in 4_results folder, respectively.
- Rerunning the data used for the analysis, bootstraps, or publication bias results.
  - To rerun the data without rerunning bootstraps or publication bias, in the masterfile, keep rerun_data as "yes" and change bootstraps and run_pub_bias to "no".
  - Adjust accordingly for only rerunning bootstraps or publication bias.
- Changing specifications:
  - Lines 73 - 151 of the masterfile determine which specifications you want for each run. Currently, the masterfile will run all seven specifications used in the paper. Here are the directions for altering these specifications:
    - Alter the mode to baseline by changing "current" to "baseline" [1].
    - Change the second metafile.do argument in each of the seven runs to alter the social cost of carbon (SCC). Currently the masterfile uses an SCC of 193, 76, 337, and 1367.
    - The third, fourth, and fifth metafile argument decides if learning by doing, include profits, and include savings is turned on or off. These can all be changed from "yes" to "no" and vice versa.
    - In the sixth argument, "all_programs" can be changed to a specific list of policies.
    - The seventh argument is the number of reps, and should always be set to zero as bootstrapping occurs in the next section of the masterfile.
    - The eighth argument is the name of the run, which can be altered as well.

If you are regenerating all the data and the bootstraps, you can expect this code to take over 72 hours to run. Once finished, the resulting dataset will be populated in the 4_results folder with

[1] Current references the 2020 specification, and baseline references the in-context specification. Most of the results reported in the paper use the current specification.

the timestamp and name of the run. The code stores the bootstrapped MVPFs and category averages in 5_graphs/figures_data with the v3 file name. Note that the figures, tables, and robustness numbers use the most recent version of the run that matches the current run name. If the name of the run is altered, the corresponding code in the figtab will need to be updated as well.

6 List of tables and programs

All main figures are produced by wrapper/figures.do, appendix figures by wrapper/appendix_figures.do, main tables by wrapper/tables.do, and appendix tables by wrapper/appendix_tables.do.

Table 1
Figure production programs

Figure #	Called Program	Output file
	Main Figures	
Figure 1	waterfalls_rep.do	waterfall_muehl_efmp_current.png
Figure 2a	waterfalls_rep.do	waterfall_hitaj_ptc_current.png
Figure 2b	wind_solar_paper.do	Fig_2b_wind_elasticities.png
Figure 3a	waterfalls_rep.do	waterfall_pless_ho_current.png
Figure 3b	wind_solar_paper.do	Fig_3b_solar_elasticities.png
Figure 4	mvpf_plots.do	mvpf_subsidies_Fig4_scc193.png
Figure 5	mvpf_plots_nudges.do	corr_plot_nudges.png
Figure 6	waterfalls_rep.do	waterfall_small_gas_lr_current.png
Figure 7	mvpf_plots.do	mvpf_subsidies_Fig7_scc193.png
Figure 8	mvpf_plots.do	mvpf_subsidies_Fig8_scc193.png
	Appendix Figures	
Appendix Figure 1a	lbd_graphs_rep.do	lbd_wind.png
Appendix Figure 1b	lbd_graphs_rep.do	lbd_solar.png
Appendix Figure 1c	lbd_graphs_rep.do	lbd_batt.png
Appendix Figure 2a	connected_externalities_driving.do	vehicle_externalities.png
Appendix Figure 2b	stacked_elec_externalities.do	Ap_Fig_2b_stacked_area.png
Appendix Figure 2c	grid_externality_region.do	Ap_Fig_2_map.png
Appendix Figure 3a	mvpf_plots_add.do	mvpf_comparison_Subsidy Robustness.png
Appendix Figure 3b	changing_grid.do	changing_grid_with_scatter.png
Appendix Figure 4	mvpf_plots.do	mvpf_intl_Ap_Fig4_split_with_CIs.png
Appendix Figure 5	bevs_non_marginal.do	Ap_Fig_5_non_marginal.png
Appendix Figure 6	contour_plot.do	Ap_Fig6_rebound.png
Appendix Figure 7a	heuristic_graphs.do	funnel_plot_threshold_5_5pct_sg.pdf
Appendix Figure 7b	heuristic_graphs.do	RD_thresh.pdf
Appendix Figure 8a	cdf_plot.do	cdf_plot.pdf
Appendix Figure 8b	cdf_plot.do	cdf_plot_AK.pdf
Appendix Figure 9	mvpf_plots.do	mvpf_subsidies_App_Fig_9_scc193.png
Appendix Figure 10a	regulations.do	reg_bars_cafe_dk_gas_panel1.png
Appendix Figure 10b	regulations.do	reg_bars_cafe_dk_gas_panel2.png
Appendix Figure 11a	regulations.do	reg_bars_cafe_as_gas_panel3.png
Appendix Figure 11b	regulations.do	reg_bars_cafe_j_gas_panel3.png
Appendix Figure 11c	regulations.do	reg_bars_rps_wind_panel3.png

Table 2
Table production programs

Table #	Called Program	Output file
	Main Tables	
Table 1	excel_MVPF_tables.do	Table1_scc193_main.xlsx
Table 2	cost_per_ton.do	Table2_CE_Table_Avg_scc193
	Appendix Tables	
Appendix Table 1		policy_details_v3.xlsx
Appendix Table 2	lbd_table_rep.do	Appendix_Table1.tex
Appendix Table 3	excel_MVPF_tables.do	App_Table3_scc193_in_context.xlsx
Appendix Table 4	ci_table.do	App_Table4_CI_with_cis_new.xlsx
Appendix Table 5	excel_MVPF_tables.do	App_Table5_scc76.xlsx
Appendix Table 6	excel_MVPF_tables.do	App_Table6_scc337_main.xlsx
Appendix Table 7	excel_MVPF_tables.do	App˙Table7˙scc1367˙main.xlsx
Appendix Table 8	excel_MVPF_tables.do	App_Table8_no_lbd.xlsx
Appendix Table 9	excel_MVPF_tables.do	App_Table9_eu_grid.xlsx
Appendix Table 10	excel_MVPF_tables.do	App_Table10_no_profits.xlsx
Appendix Table 11	excel_MVPF_tables.do	App_Table11_with_savings.xlsx
Appendix Table 12	excel_ce_lbd_tables.do	Table12_CE_Table_All_Policies_with_LBD_scc193
Appendix Table 13	excel_ce_lbd_tables.do	Table13_CE_Table_All_Policies_no_LBD_193
Appendix Table 14	excel_ce_lbd_tables.do	Table14_CE_Table_DWL_193
Appendix Table 15	connected_externalities_driving.do	Appendix Table Gas Externalities.xlsx

References

Isaiah Andrews and Maximilian Kasy, “Identification of and Correction for Publication Bias,” *American Economic Review*, vol. 109, no. 8, pp. 2766–2794, August 2019.

Cassandra Cole, Michael Droste, Christopher Knittel, Shanjun Li, and James H. Stock, “Policies for Electrifying the Light-Duty Vehicle Fleet in the United States,” *AEA Papers and Proceedings*, vol. 113, pp. 316–322, May 2023.